



Securing Agentic AI at Scale

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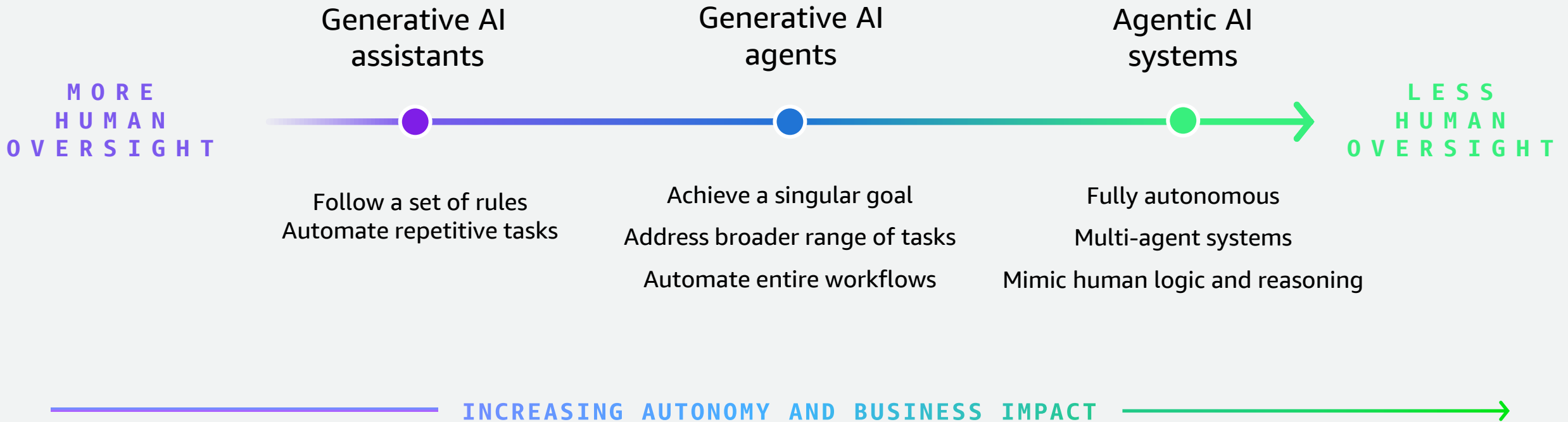
Agenda

- Agentic AI Security and Threat Modeling
- Amazon Bedrock AgentCore Identity
- Demo
- Securing Agentic AI Applications with AWS
- Summary, Resources & Call to Action
- Exit Survey + Q&A

What country are you joining from today?

Agentic AI Security and Threat Modeling

The evolution into Agentic AI



Agentic use cases across industries

AUTOMOTIVE

- AI-powered development and validation
- Intelligent contact center
- Supply chain optimization

ENERGY & UTILITIES

- Energy trading
- Renewable energy planning
- Upstream production optimization

FINANCIAL SERVICES

- Insurance underwriting optimization
- Intelligent customer experience
- Investment research assistant

GAMES

- AI for game production
- Code assistance for game development
- In-game AI

MANUFACTURING

- Research intelligence and discovery
- Maintenance management
- Supply chain optimization

TELECOM

- Customer experience
- Enterprise productivity
- Network automation

MEDIA & ENTERTAINMENT

- Multi-agent collaboration for video understanding
- Media planning and campaign optimization
- Multi-agent collaboration for advertising sales data

RETAIL & CONSUMER GOODS

- Agentic shopping assistants
- Supply chain reallocation
- Pricing agents

HEALTHCARE & LIFE SCIENCES

- Clinical workflow automation
- Prior authorization automation
- Patient and member engagement



AWS Agentic AI portfolio

APPLICATIONS

Kiro

Amazon Quick Suite

AWS Transform

Amazon Connect

AWS Marketplace

AI & AGENT DEVELOPMENT SOFTWARE & SERVICES

SDKS FOR AGENTS

Vertically Integrated

Nova Act

Flexible/OSS

Strands Agents

AMAZON BEDROCK

Models

Amazon Nova

3P Models

Capabilities

Optimization

Guardrails

Customization

AgentCore

Runtime

1P Tools

Gateway

Identity

Memory

Observability

Knowledge Bases

INFRASTRUCTURE

AMAZON SAGEMAKER

Custom Model Building

Model building

Model training

MLOps

Deployment

Hyperpod

Data Foundation for AI

Data Processing

Governance

Storage and Databases

Vector Search

AI COMPUTE

Trainium

Inferentia

GPUs

INTERFACES & PROTOCOLS
(MCP/A2A)

SECURITY & POLICIES



How to think about

agentic AI security

COMPARED TO GENERATIVE AI WORKLOADS



Poll

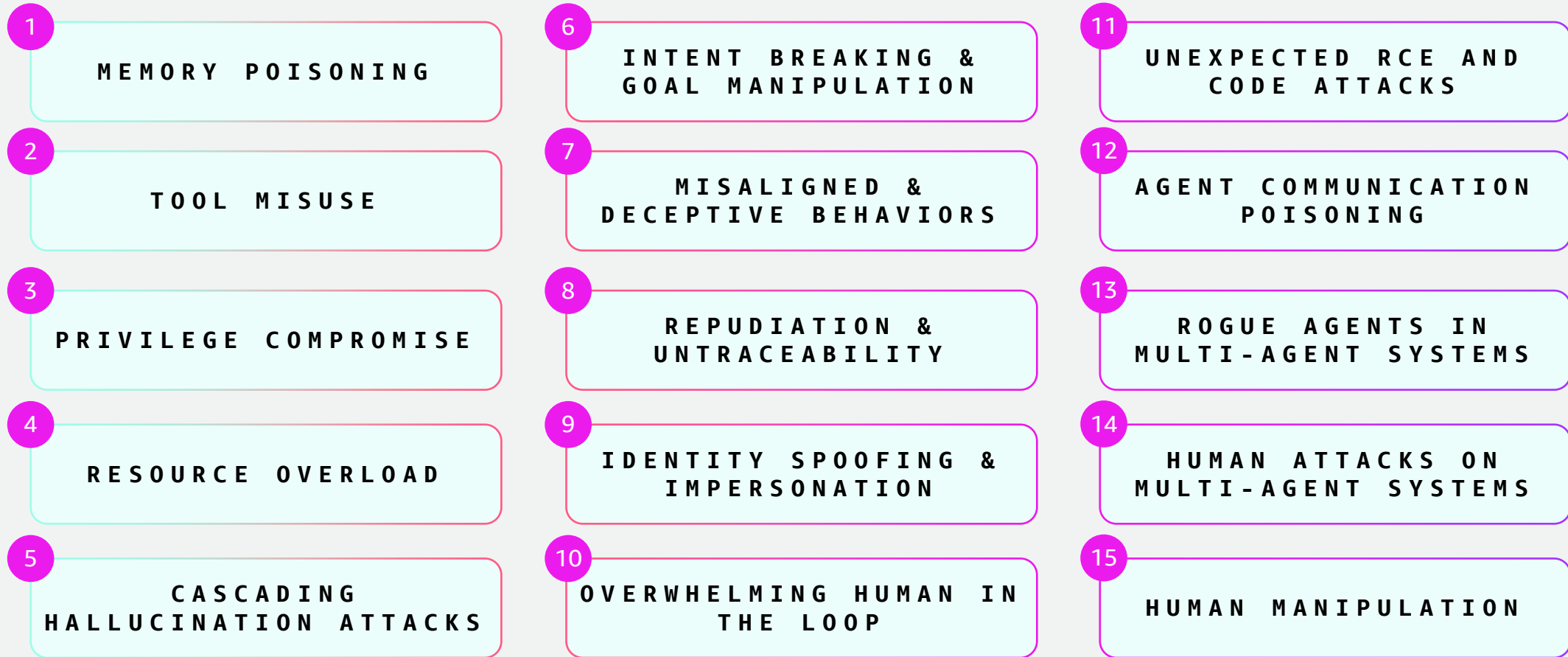
What's the #1 security risk with AI agents?

How should you think
about

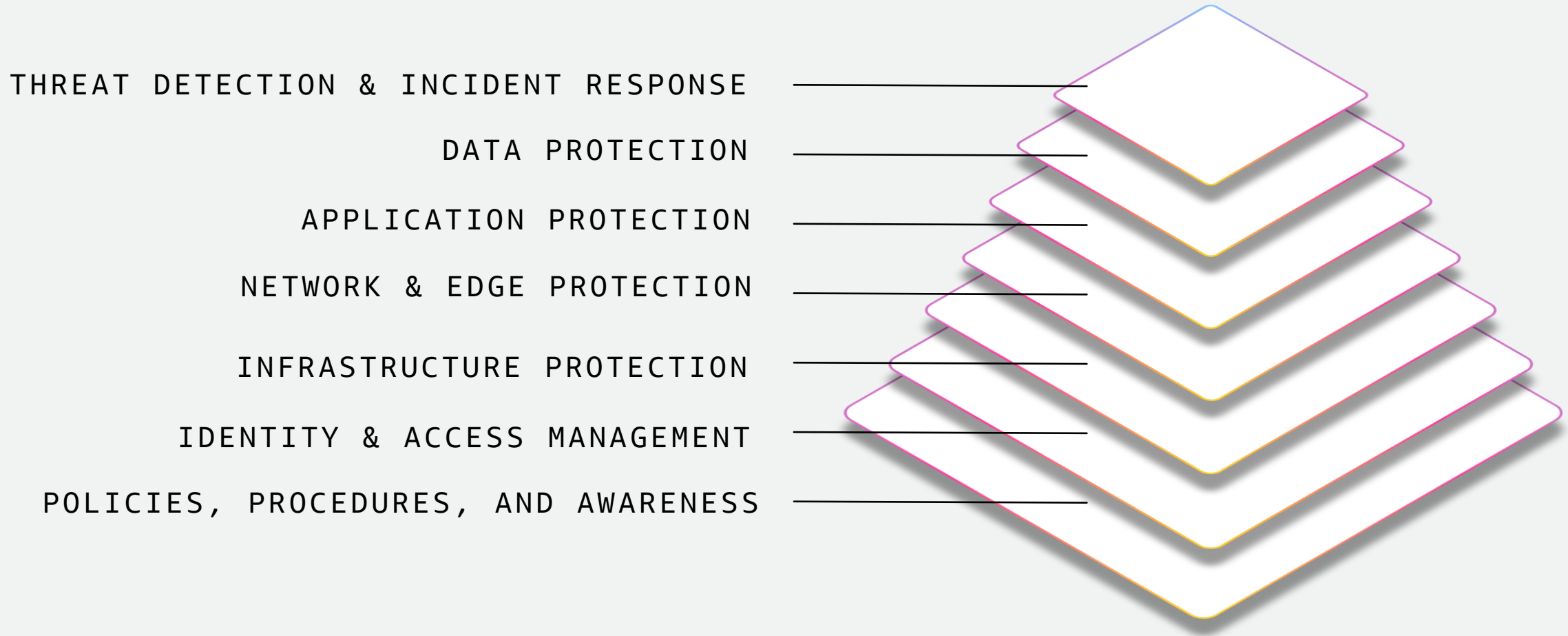
threat modeling

WITH AI AGENTS

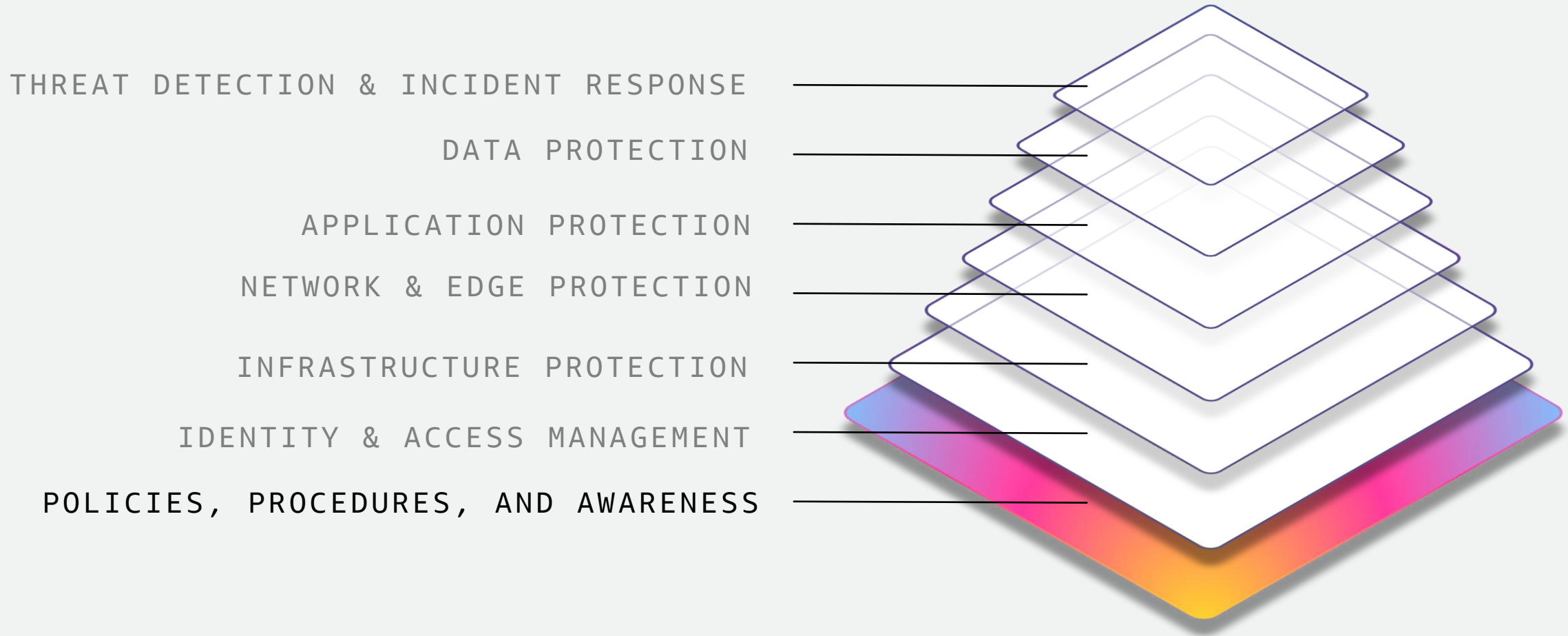
OWASP Agentic AI – Threats and Mitigations



Defense in Depth Security



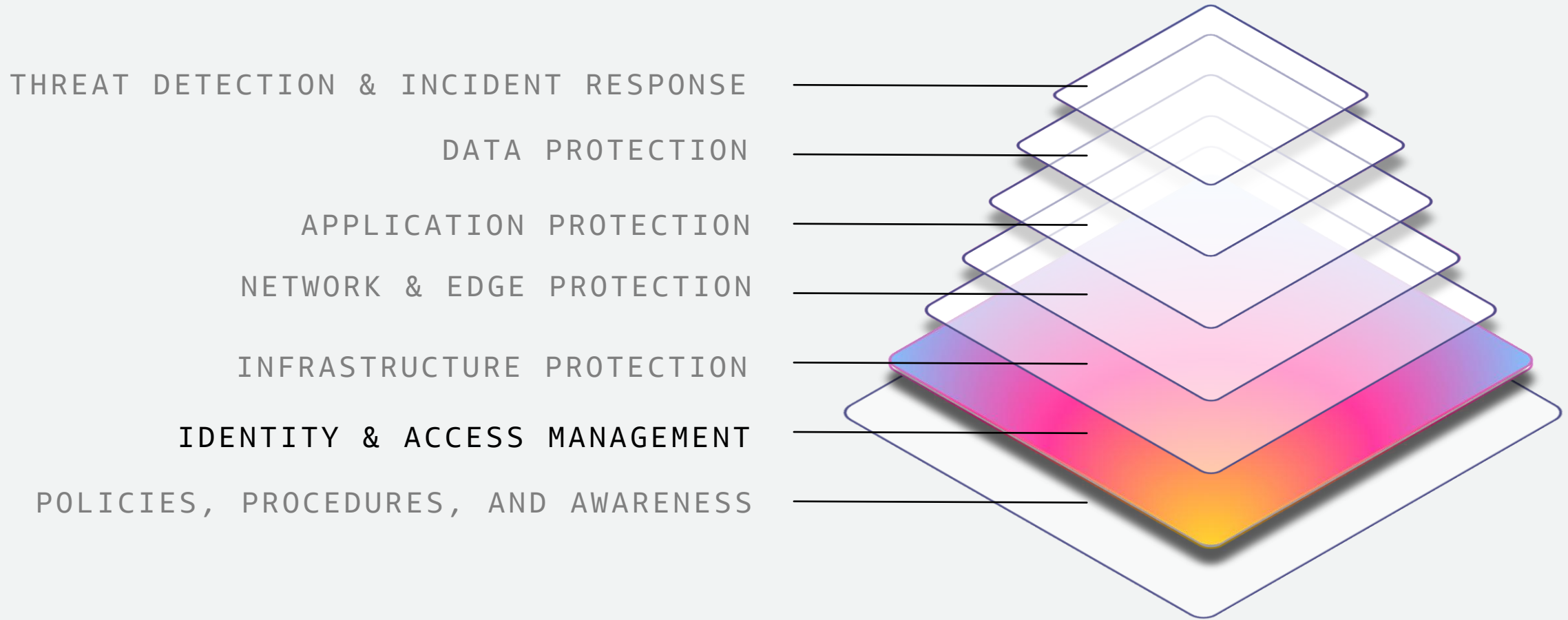
Defense in Depth Security

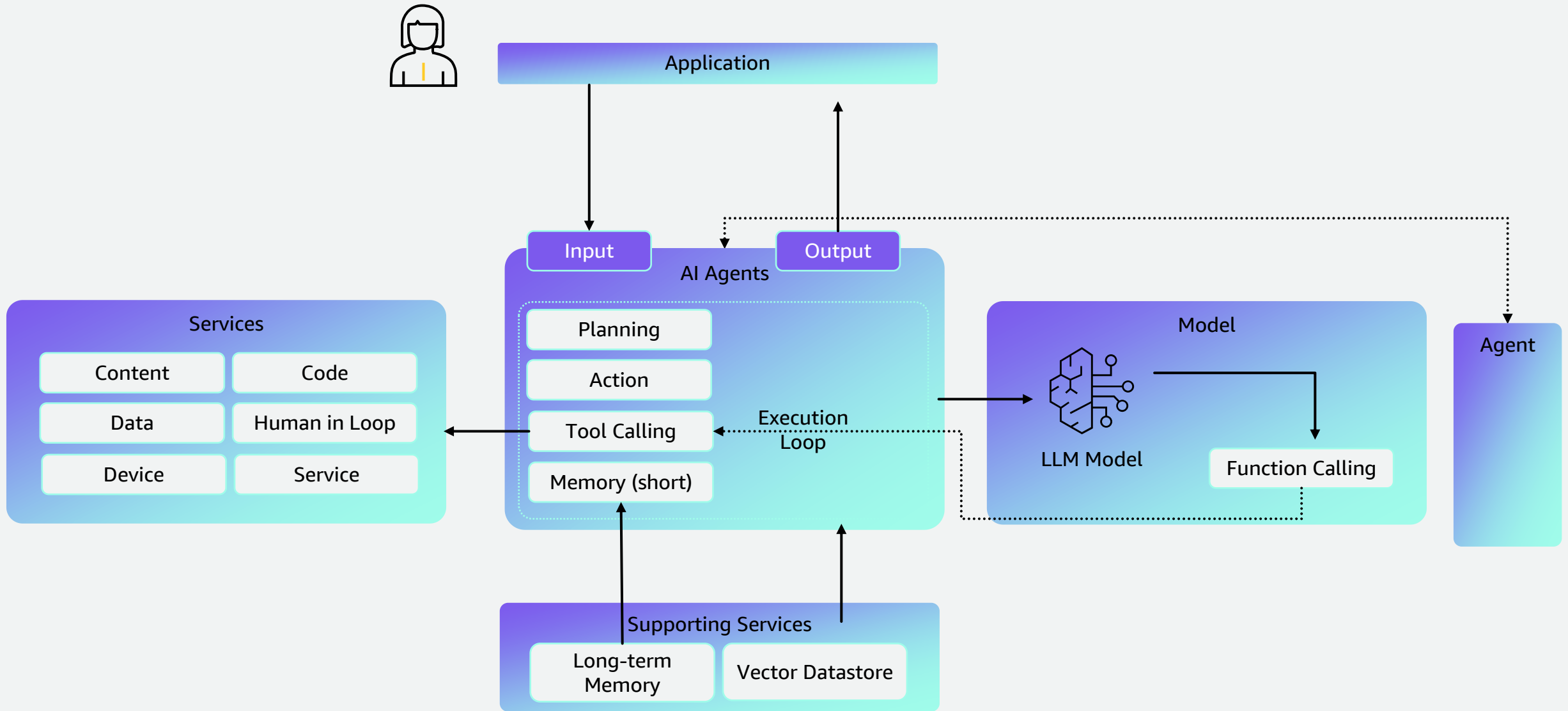


Policies, Procedures, and Awareness

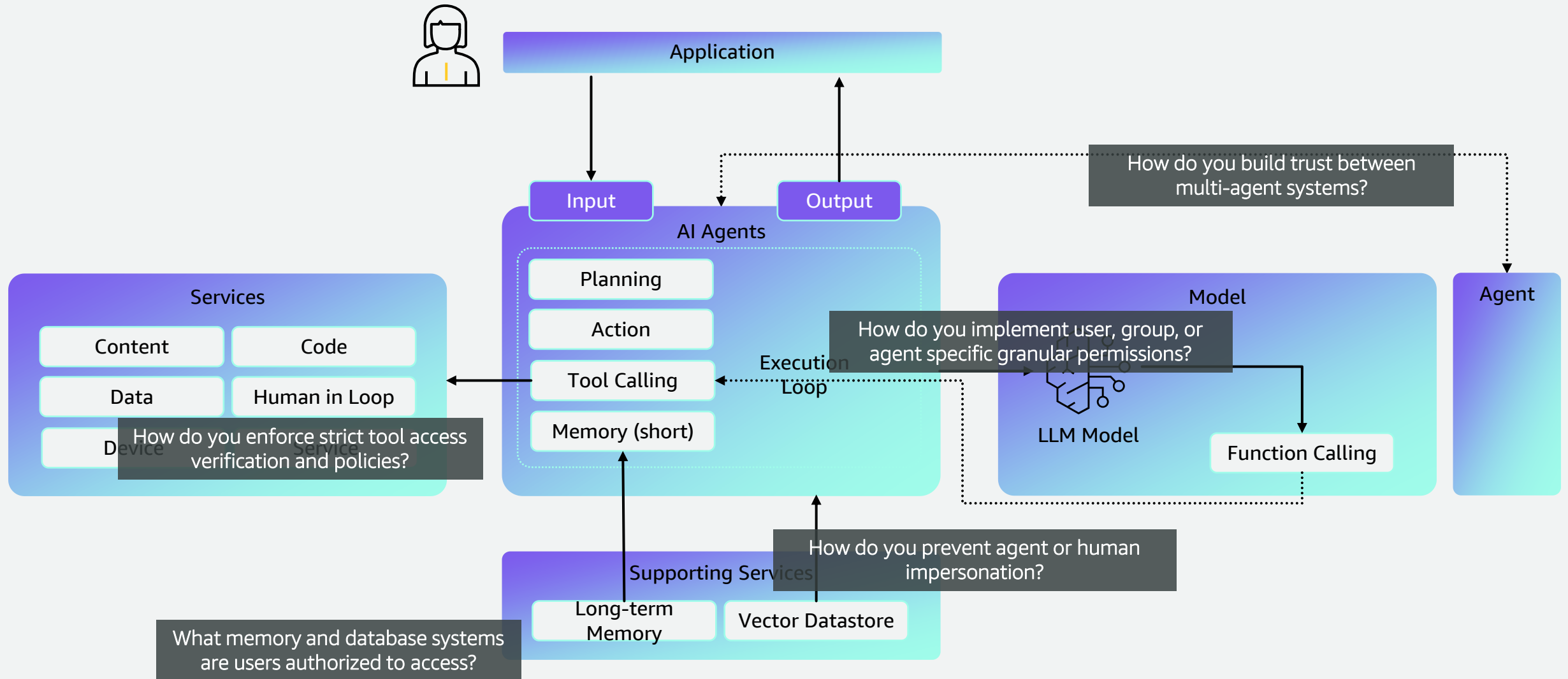


Defense in Depth Security





Identity & Access Management



Agentic AI Identity on AWS

AMAZON COGNITO



Frictionless and security-focused customer identity & access management and M2M authentication

AMAZON BEDROCK AGENTCORE IDENTITY



Identity and credential management service for AI Agents

AWS IAM



Manage access to AWS resources

Poll

**Your AI agent is like a new employee.
What do you give them first?**

Amazon Bedrock AgentCore Identity

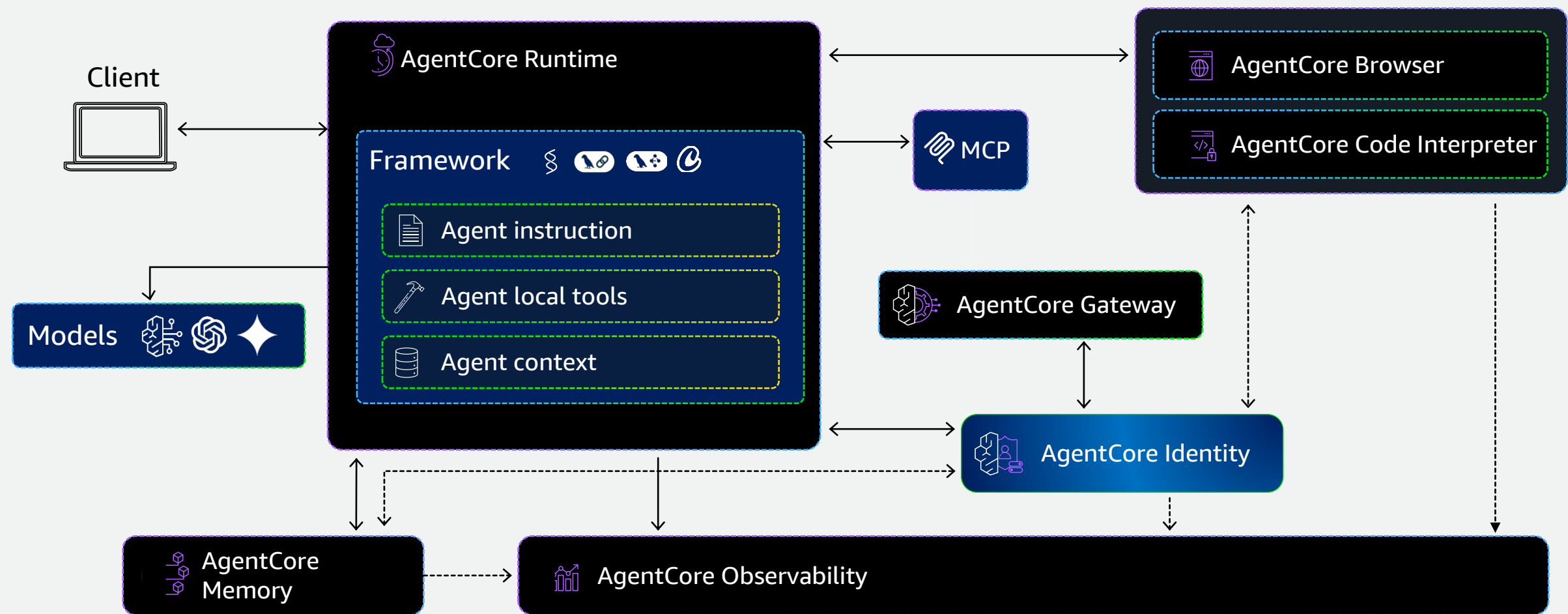




Amazon Bedrock AgentCore

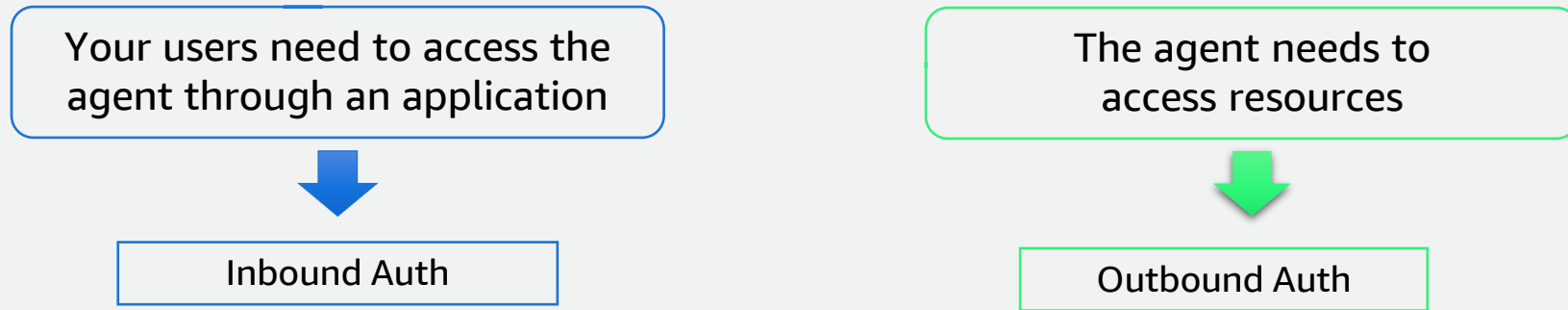
Build, deploy, and operate highly capable agents securely at scale using any framework and model – no infrastructure management required

Amazon Bedrock AgentCore

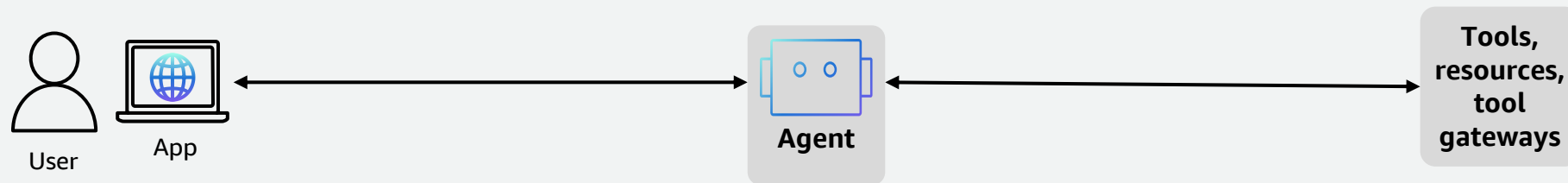


Agentic AI Auth Basics

As you're building an AI agent...

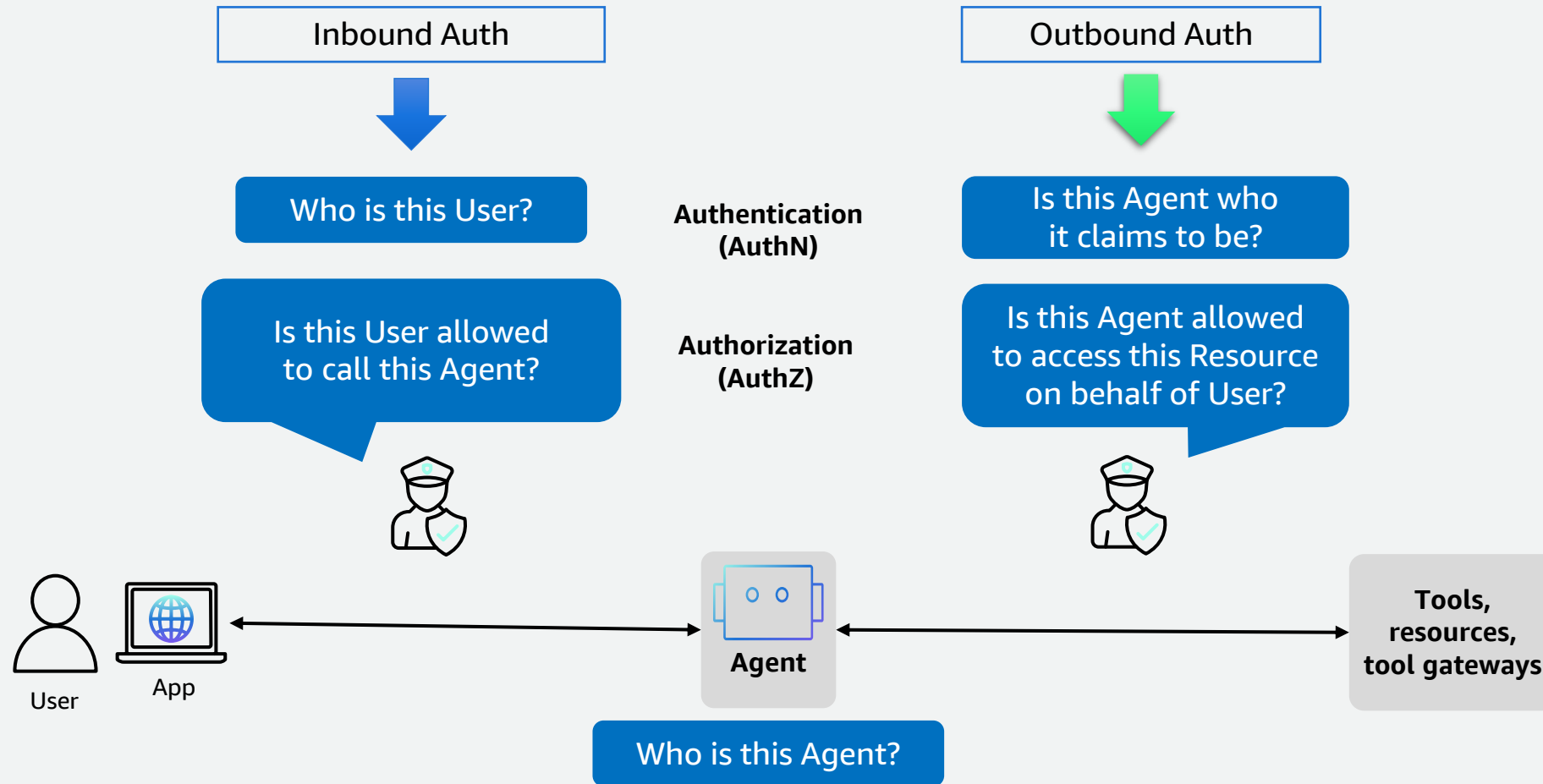


Bedrock AgentCore Identity can help you with both...



...and we can also host your agent code

Agentic AI Auth Basics (cont.)

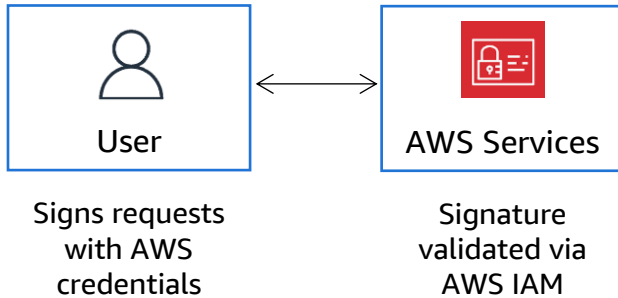


Common Auth Patterns in a Nutshell

AWS Credentials (Sigv4)

- Act as the authenticated user
- For direct user-to-service authentication
- User signs requests with their own AWS credentials
- No token exchange or delegation required

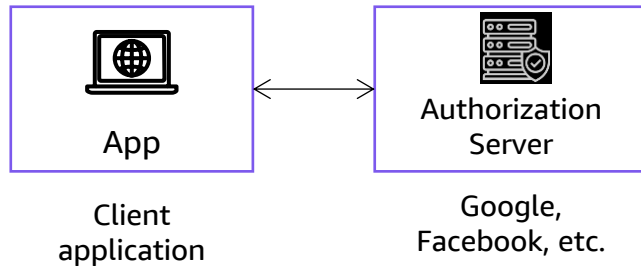
Example: User authenticates to access a bucket in Amazon S3, or to invoke an agent in Bedrock AgentCore Runtime.



OAuth - 2-Legged OAuth (2LO) (also known as M2M)

- Act as the service
- For server-to-server communication
- Client app directly auth with server

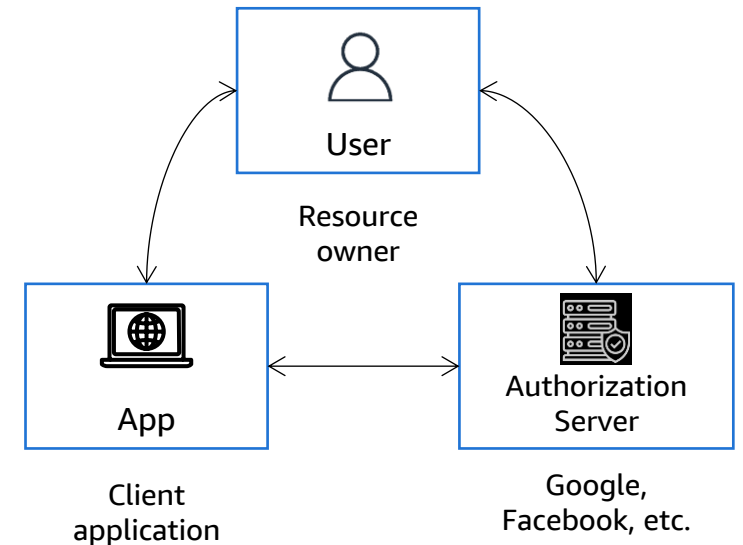
Example: Agent accesses external APIs (company knowledge base, weather service) using its own OAuth client credentials.



OAuth - 3-Legged OAuth (3LO)

- Act as the user
- For user auth app-to-server communication
- User consent required via auth server
- App access resources on behalf of the user

Example: Agent accesses external APIs (Google, Slack) on user's behalf



Quiz

True/False - It's okay to give your agent admin access "just in case"



AgentCore Identity: Enterprise agent identity and access management

Secure, delegated access for AI agents across enterprise services



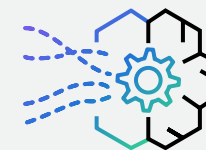
- Enables AI agents to securely access AWS resources and third-party tools such as GitHub, Google, Salesforce and Slack
- Enhanced access controls with just-enough access, secure permissions delegation, and identity-aware authorization

Build streamlined AI agent experiences



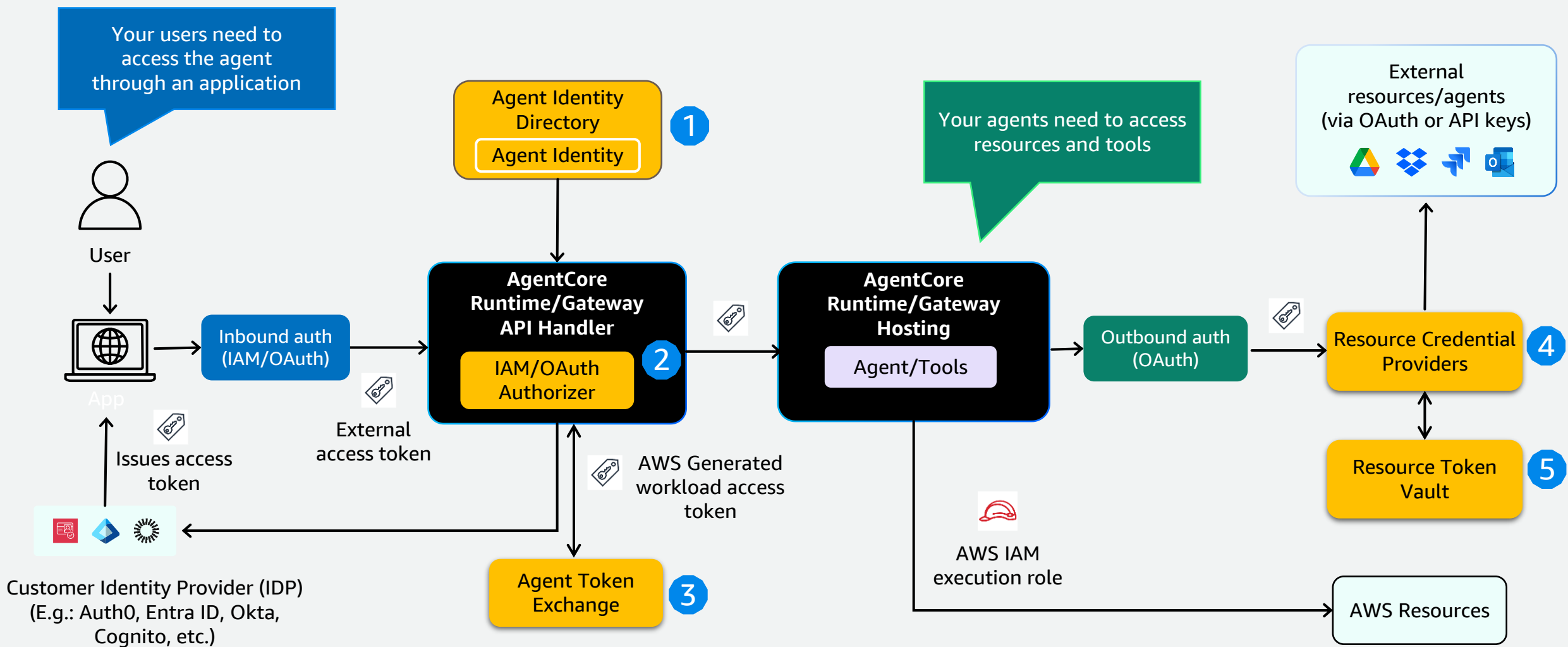
- Minimizes consent fatigue with a secure token vault
- Streamlines authentication flows
- Delivers frictionless user experiences for all your agent-powered interactions

Accelerate AI agent development

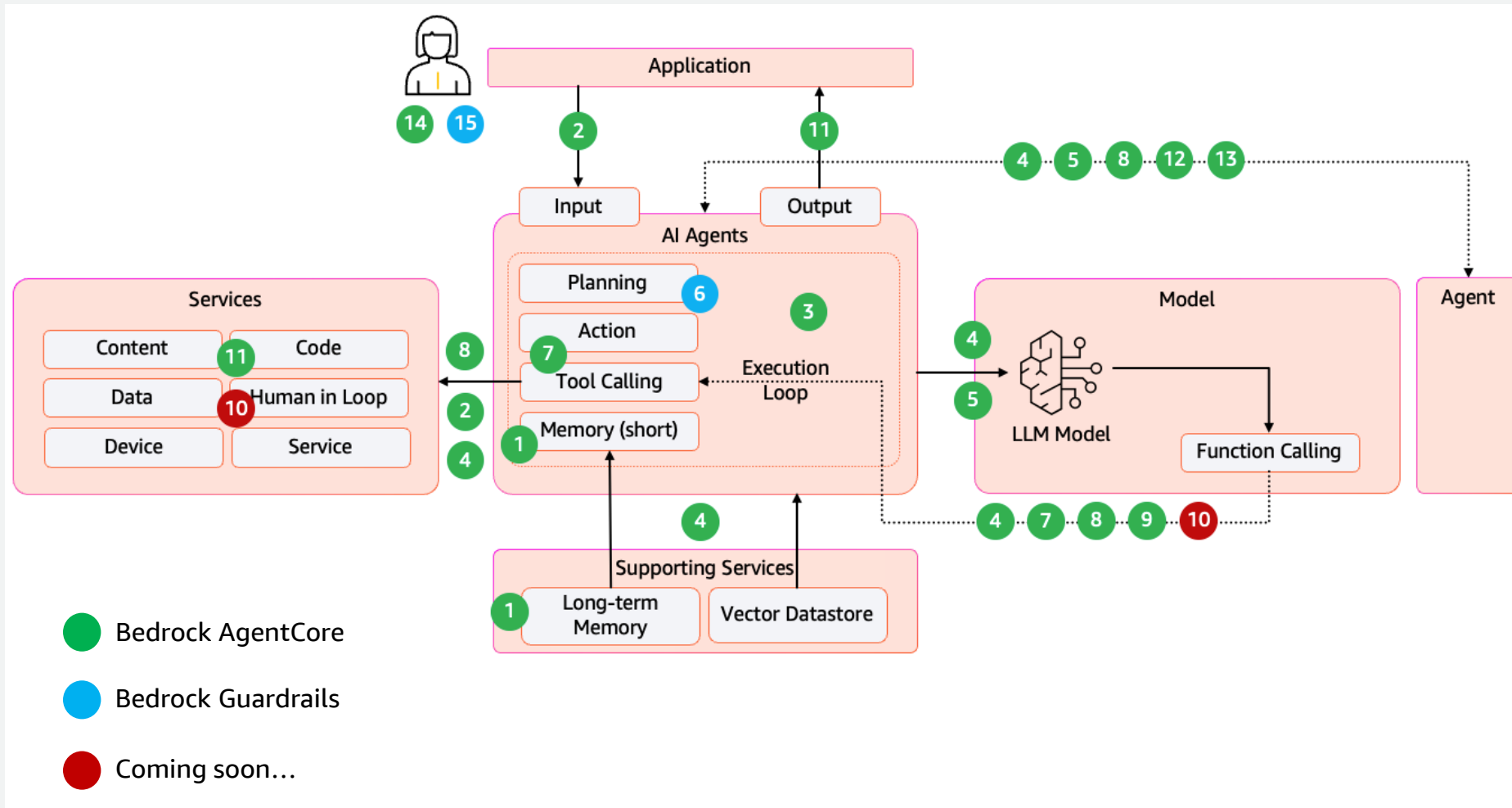


- Preserves existing identity systems such as Okta, Microsoft Entra ID, or Amazon Cognito
- Lowers custom development efforts without need for migrating users or rebuilding authentication flows

Auth with AgentCore Identity



OWASP Agentic AI – Threats and Mitigations using Bedrock AgentCore



- 1 **MEMORY POISONING**
- 2 **TOOL MISUSE**
- 3 **PRIVILEGE COMPROMISE**
- 4 **RESOURCE OVERLOAD**
- 5 **CASCADING HALLUCINATION
ATTACKS**
- 6 **INTENT BREAKING & GOAL
MANIPULATION**
- 7 **MISALIGNED & DECEPTIVE
BEHAVIORS**
- 8 **REPUDIATION & UNTRACEABILITY**
- 9 **IDENTITY SPOOFING &
IMPERSONATION**
- 10 **OVERWHELMING
HUMAN IN THE LOOP**
- 11 **UNEXPECTED RCE
AND CODE ATTACKS**
- 12 **AGENT COMMUNICATION
POISONING**
- 13 **ROGUE AGENTS IN MULTI-AGENT
SYSTEMS**
- 14 **HUMAN ATTACKS ON MULTI-AGENT
SYSTEMS**
- 15 **HUMAN MANIPULATION**

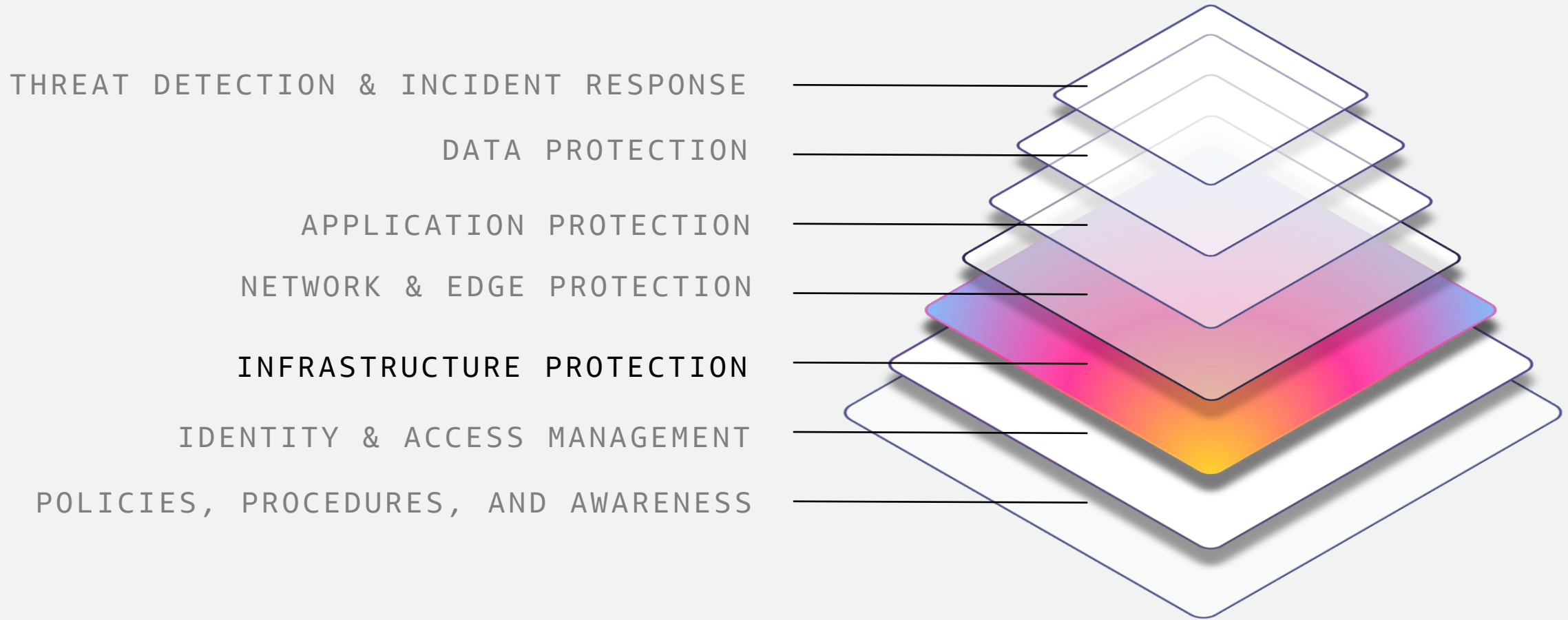


Demo

Securing Agentic AI Applications with AWS



Defense in Depth Security



Defense in Depth Security

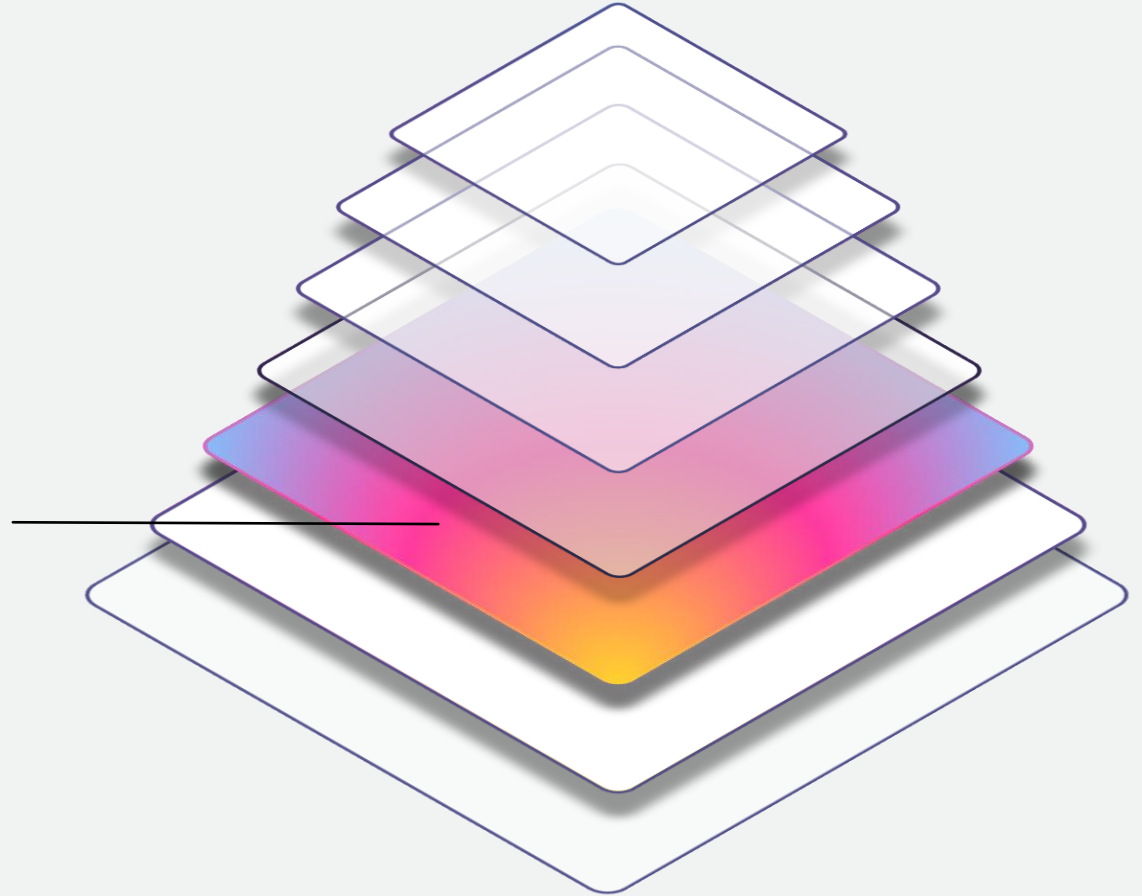


VPC Security
Groups



AWS Network
Firewall

INFRASTRUCTURE PROTECTION



Defense in Depth Security

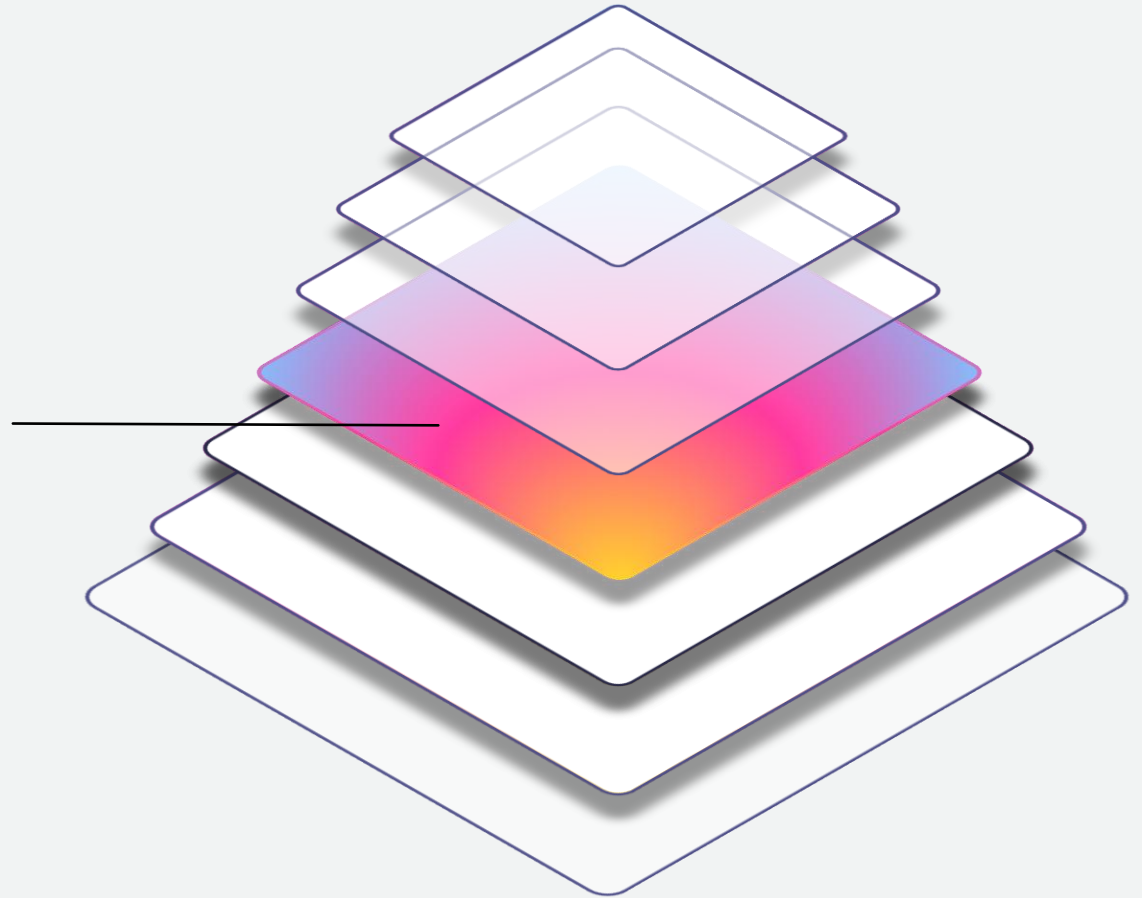


AWS WAF

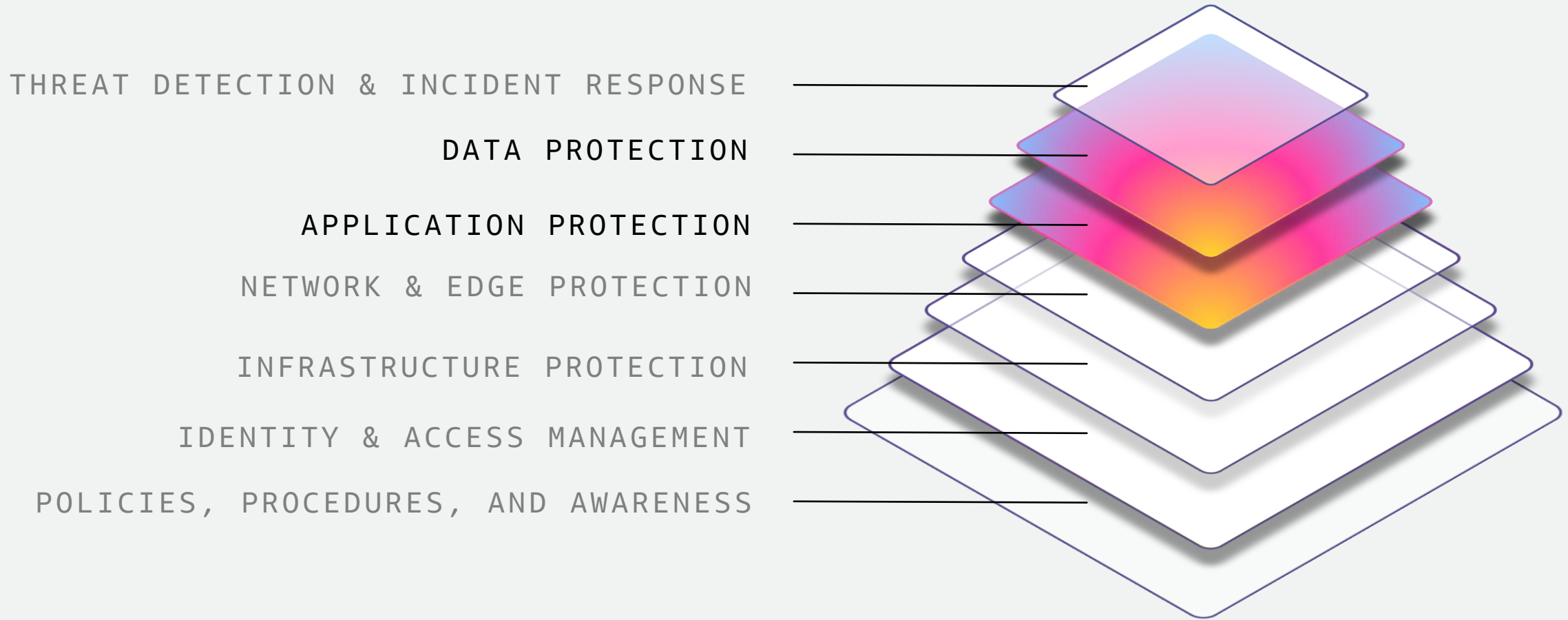


AWS Shield
Advanced

NETWORK & EDGE PROTECTION

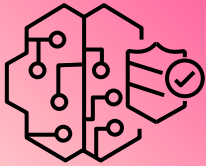


Defense in Depth Security



Application and Data Protection

GUARDRAILS



Evaluate prompts and
model responses

VALIDATION



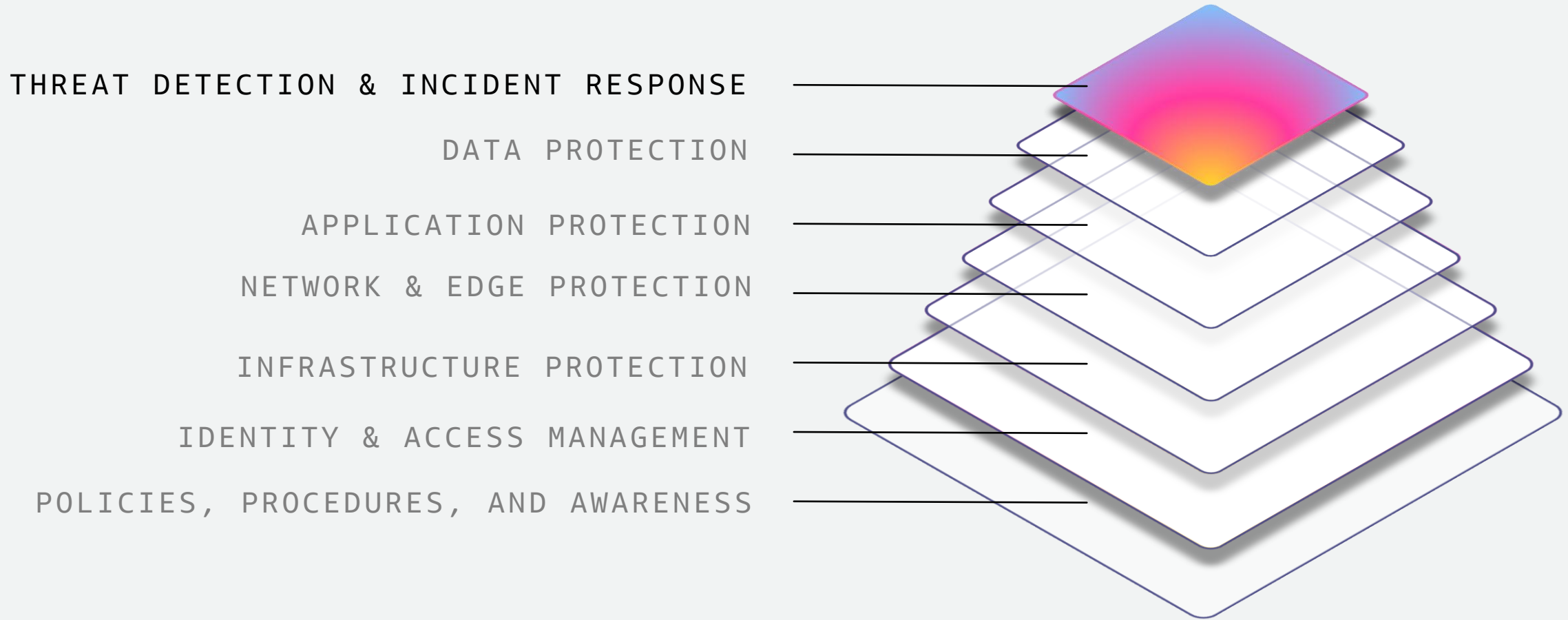
Input and output
validation for prompts

AUTOMATED REASONING

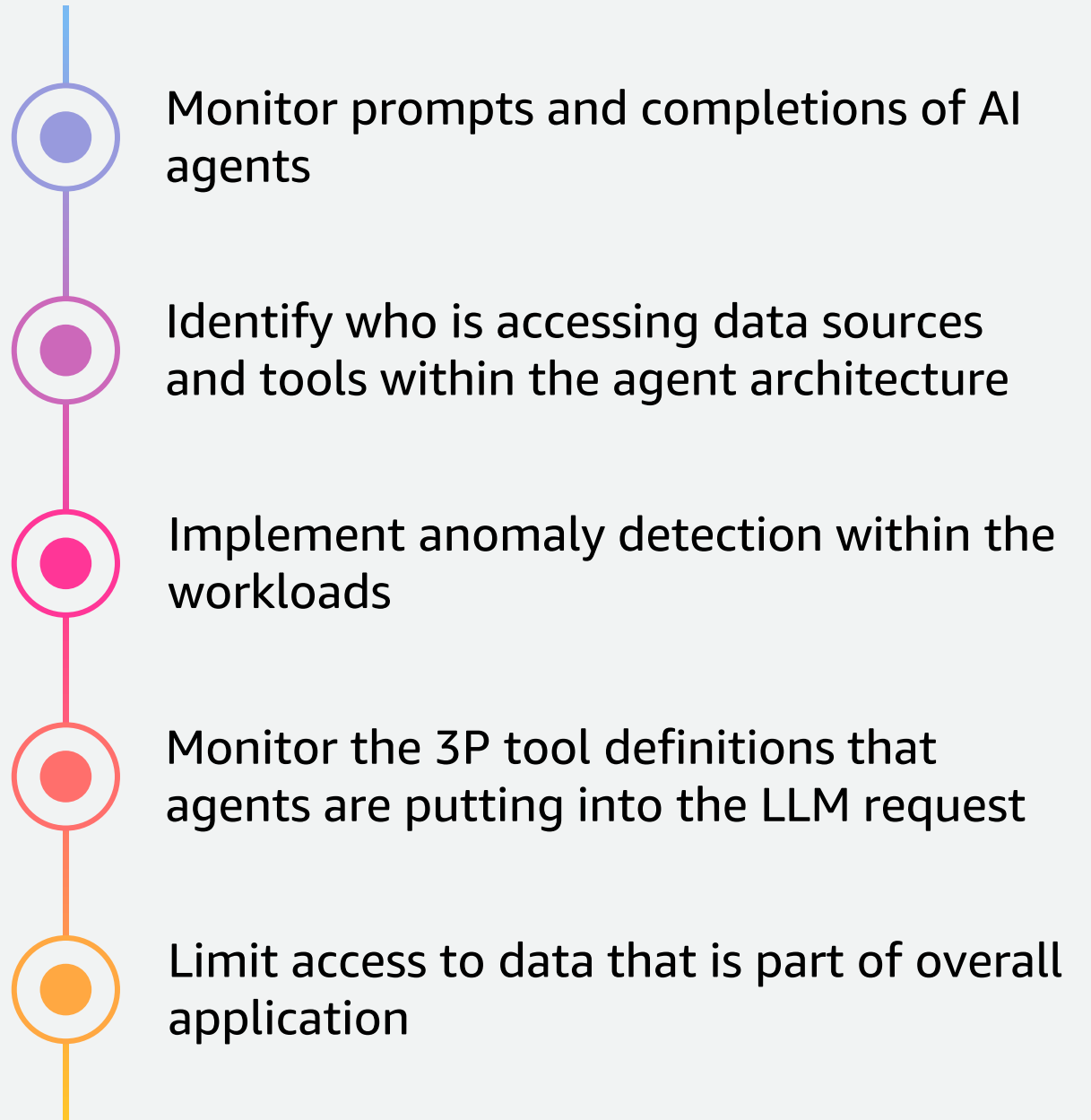


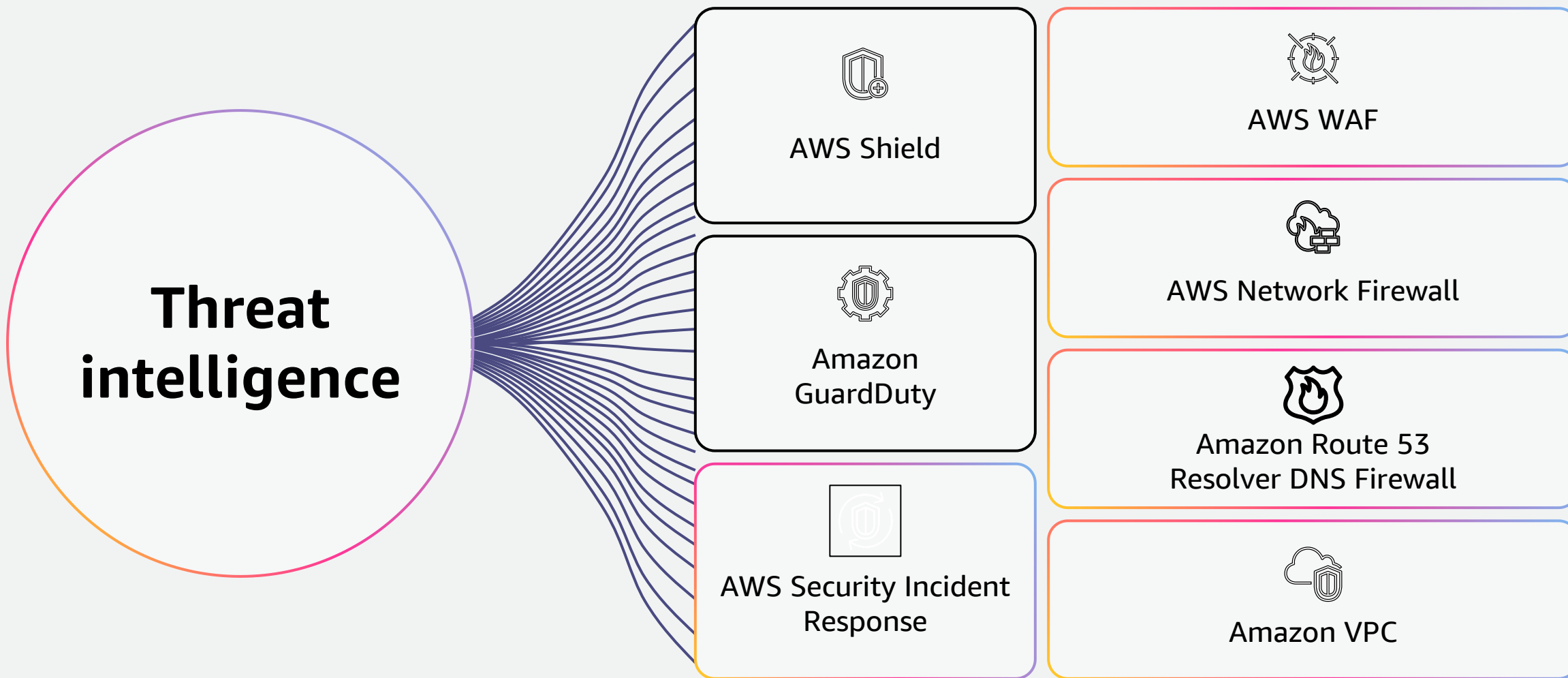
Detect factual
errors with
verifiable reasoning

Defense in Depth Security



Threat Detection and Response for AI Agents





Quiz

**Complete this sentence:
"An agent without proper security is
like..."**

Summary



Agentic AI requires evolved security thinking - it's not just about data, it's about autonomous actions

AWS provides integrated security services across all protection layers

Strong policies and procedures form the foundation of technical controls

AgentCore Identity enables secure enterprise AI agent deployment at scale

Dual authentication patterns address unique autonomous agent security challenges effectively

Resources

**Securing
Generative AI**



AWS Security



**AgentCore Identity
AWS Show & Tell**



Poll

After this session, how confident are you in implementing secure agent identity?

Call To Action



Get Started : Journey to become an Agentic AI Specialist



Join Upcoming Sessions!

Scan QR code to register now:



**Build Intelligent AI Agents Using Strands SDK and
Model Context Protocol (MCP)**

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